

Karthik Bhattaram

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EDUCATION

University of California, Santa Barbara

Master of Science in Computer Science

Santa Barbara

Sept. 2026 to June 2027

- Concentration: Machine Learning, Network Security, Systems

University of California, Santa Barbara

Bachelor of Science in Computer Science

Santa Barbara

Sept. 2022 to June 2026

- Relevant Coursework: Machine Learning for Networking, Computer Security, Data Structures and Algorithms
- Honors: Regents Scholarship, College of Engineering Honors, USACO Silver Division

EXPERIENCE

Software Engineering Intern

Arista Networks

June 2025 – Sept. 2025

Santa Clara, California

- Created utilities to decode hardware registers specifying pause storm information in Ethernet ports
- Implemented a **PFC watchdog** that relayed the utility's output to a network switch's traffic manager

Machine Learning Researcher

University of California, Santa Barbara

Sept. 2024 – June 2025

Santa Barbara

- Developed pipelines to automate common network measurement efforts
- Curated network traffic datasets for a RAG-based **large language model**

Software Engineering Intern

Cadence Design Systems / OpenEye Scientific

July 2024 – Sept. 2024

Santa Fe, New Mexico

- Wrote **AWS**-based package to evaluate database compression algorithms
- Updated **CUDA C++** software using new language features to match competing solutions

Hardware Architecture Intern

Cadence Design Systems

June 2023 – Sept. 2023

San Jose, California

- Developed an automation tool to produce memory wrappers of various dimensions and architectures
- Collected **System Verilog** files defining memory modules to input to the tool

Software Engineering Intern

Axiado Corporation

June 2021 – Aug. 2022

San Jose, California

- Developed a package to create **network packet flows** based on IPv6, ARP, ICMP, etc. in simulation
- Deployed and expanded test suites of a hardware firewall's packet parser module

PROJECTS

Weather Predictor | *Pandas, Numpy, Arparse, Excel, Bash*

March 2024 – May 2024

- Parsed Excel sheets and inputted their data into a Pandas dataframe
- Used a Naïve Bayes classifier implemented with Numpy
- Achieved **best-in-class accuracy** of 68.5% among 96 students
- Wrote a Bash automation script to run the training program

TECHNICAL SKILLS

- **Certificates:** Enterprise Security Professional, Data Science
- **Networking Technologies:** Scapy, Wireshark, TCP/IP
- **Languages:** Python, C/C++, Java, JavaScript, Verilog
- **Developer Tools:** Git, Visual Studio Code, Vim, Emacs
- **Libraries:** CUDA, Pandas, NumPy, PyTest, Black, Matplotlib